THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–4: **Large Numbers, Arithmetic Expressions, A Peek Beyond the Point, Expressions using Letter-Numbers** | Class 7 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 6,04,800	1 day = 86,400 seconds.
2	**(b)** 15	Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction.
3	**(a)** 0.024	Multiply two at a time and count the decimal places carefully.
4	**(a)** 21	Square first, then multiply.
5	**(c)** 7	Count zeros in the product.
6	**(d)** 15	Sum first, then double, then subtract.
7	**(d)** 48	Divide the larger number by the smaller.
8	**(b)** $2x + 13$	Expand each bracket, watching the signs.
9	**(d)** 25	Divide by 1,00,00,000.
10	**(c)** -4	Evaluate both products.
11	**(c)** 7	Divide the part by the fraction it represents.
12	**(d)** $16x + 6$	Perimeter $= 2\times$(length + breadth).
13	**(c)** 48	Divide.
14	**(d)** 2277	Distributive property.
15	**(b)** 1.7	Work out the bracket first.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	1 day = 86,400 seconds (1); $7 \times 86400 = 6,04,800$; international: 604,800 (1).	1+1
17	Innermost brackets; Next level (1); Final operations; Value $= 15$ (1).	1+1
18	The missing number is the product divided by the known factor (1); $\square = 0.162 \div 0.6 = 0.27$ (1).	1+1
19	Expand: $7x - 14 - 3x + 15$ (1); Answer: $4x + 1$ (1).	1+1
20	Compare digit counts and then place values (1); Greater: 98,76,543; difference $9,000$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Total $= 2,50,000 \times 48 = 1,20,00,000$ (1); Words: one crore twenty lakh (1); Nearest crore: $1,00,00,000$ (1).	1+1+1
22	Full sets $= 24 \div 6 = 4$; Discount $= 4 \times 20 = 80$ (1); Total $= 24 \times 35 + 30 \times 12 - 80 = 1120$ (1); Check with units (₹) (1).	1+1+1
23	$2.5 \times 0.4 = 1$ (1); $1.6 \div 0.4 = 4$ (1); Sum $= 5$ (1).	1+1+1
24	$P = 2[(5x + 4) + (3x - 1)]$; $= 2(8x + 3)$ (1); $= 16x + 6$ (1); At $x = 6$: 102 cm (1).	1+1+1
	OR: Compare numbers digit by digit (1); Descending: 12,43,567, 12,34,765, 12,34,567 (1); Difference $= 9,000$ (1).	1+1+1
25	First $= 180$ (1); Second $= (12 + 2) \times (15 - 2) = 182$ (1); Difference $= 2$ (1).	1+1+1
	OR: They separate by $3.5 + 4.5 = 8$ km every hour (1); After 2.5 hours: $8 \times 2.5 = 20$ km (1); Time $= 24 \div 8 = 3$ hours (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Monthly $= 2,50,000 \times 180 = 4,50,00,000$ (1); Yearly $= 54,00,00,000$ (1); Words: fifty-four crore (1); Nearest crore: $54,00,00,000$ (1); Show working with place values (1).	1+1+1+1+1
	OR: Income $= 60 \times 3 \times 10 + 40 \times 2 \times 25$ (1); $= 3800$ (1); Expenses $= 800 + 400 = 1200$ (1); Profit $= 2600$ (1); Expression uses brackets correctly (1).	1+1+1+1+1
27	Net gain per minute $= 7.5 - 2.5 = 5$ litres (2); Time $= 60 \div 5 = 12$ minutes (3).	2+3

	OR: (i) $4(15)+3=63$ (1); (ii) Figure 1 $=7$ and figure 2 $=11$ (1); Total $=18$ (1); (iii) $63-59=4$ more tiles (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–4: **Large Numbers, Arithmetic Expressions, A Peek Beyond the Point, Expressions using Letter-Numbers** | Class 7 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 55	Substitute, multiply, then halve.
2	**(c)** 43,211	Round up to the next lakh.
3	**(a)** 100	Distributive property.
4	**(c)** 0.95	Find how far each number is from 1.
5	**(a)** $2n + 4$	Test the expressions with small values of n.
6	**(b)** 125	Divide by a lakh.
7	**(b)** 130	Add the decreasing distances.
8	**(b)** 2.65	Subtract the smaller number from the larger.
9	**(b)** 22	Substitute carefully; xy means $x \times y$.
10	**(d)** 10	Multiply and count.
11	**(d)** 3	Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction.
12	**(a)** $\frac{1}{8}$	Write over 1000, 100 or 10 and reduce.
13	**(d)** 23	Apply the rule to the input.
14	**(c)** 1800	1 crore = 100 lakh.
15	**(a)** 24	Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$= 37 \times (48 + 52)$ (1); $= 37 \times 100 = 3700$ (1).	1+1
17	Largest: 85.42 (digits in decreasing order); Smallest: 24.58 (digits in increasing order) (1); Difference $= 85.42 - 24.58 = 60.84$ (1).	1+1
18	$2n + 2(n + 1) + 2(n + 2)$ (1); $= 6n + 6 = 6(n + 1)$ (1).	1+1
19	1 lakh more $= 1,24,45,678$ (1); 1 crore less: $24,45,678$ (1).	1+1
20	First $= 150$; Second $= 154$ (1); First is smaller (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Numbers: $n, n + 1, \ldots, n + 2$ (1); Sum $= 3n + 3$ (1); At $n = 7$: 24 (1).	1+1+1
22	Distance $= 300000 \times 8 = 24,00,000$ km (1); In words: twenty-four lakh (2).	1+2
23	Outer area $= (20 + 2 \times 2) \times (12 + 2 \times 2) = 384$ (1); Garden area $= 20 \times 12 = 240$ (1); Path area $= 144$ m^2 (1).	1+1+1
24	Price per kg $= 135 \div 2.5 = 54$ (1); Cost of 4.5 kg $= 54 \times 4.5 = 243$ (2).	1+2
	OR: A: $20 + 15k$ and B: $60 + 10k$ (1); For 6 km: A $= 110$ and B $= 120$ (1); Taxi A is cheaper by ₹10 (1).	1+1+1
25	Total $= 1,25,000 \times 96 = 1,20,00,000$ (1); Words: one crore twenty lakh (1); Nearest crore: $1,00,00,000$ (1).	1+1+1
	OR: Full sets $= 30 \div 5 = 6$; Discount $= 6 \times 25 = 150$ (1); Total $= 30 \times 40 + 25 \times 15 - 150 = 1425$ (1); Check with units (₹) (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Larger $= (1,25,00,000 + 15,00,000) \div 2 = 70,00,000$ (1); Smaller $= 55,00,000$ (1); Words: seventy lakh; fifty-five lakh (1); Check: sum and difference match (2).	1+1+1+2
	OR: First $= 21$ (1); Second $= -3$ (1); Larger: first (1); Difference 24 (2).	1+1+1+2
27	Total cost $= 180 + 500 = 680$ (1); Total weight $= 4$ kg (2); Cost per kg $= 680 \div 4 = 170$ (2).	1+2+2

	OR: Brother's money $= 2x + 15$ (1); Total $= x + (2x + 15) = 3x + 15$ (1); At $x = 40$: brother $=$ ₹95 (1); Total $=$ ₹135 (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–4: **Large Numbers, Arithmetic Expressions, A Peek Beyond the Point, Expressions using Letter-Numbers** | Class 7 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** $\frac{1}{2}$	Convert every number to a decimal and compare.
2	**(c)** 77	$x^2 = x \times x$.
3	**(c)** 25,92,000	1 day = 86,400 seconds.
4	**(a)** 82	Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction.
5	**(a)** 0.18	Multiply two at a time and count the decimal places carefully.
6	**(a)** 13	Square first, then multiply.
7	**(d)** 7	Count zeros in the product.
8	**(d)** 25	Sum first, then double, then subtract.
9	**(d)** 24	Divide the larger number by the smaller.
10	**(b)** $3x + 29$	Expand each bracket, watching the signs.
11	**(d)** 4.5	Divide by 1,00,00,000.
12	**(c)** -6	Evaluate both products.
13	**(a)** 8	Divide the part by the fraction it represents.
14	**(c)** $18x + 6$	Perimeter $= 2\times$(length $+$ breadth).
15	**(c)** 36	Divide.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Add: $0.3 + 0.4 = 0.7$ (1); Halve: $0.7 \div 2 = 0.35$ (1).	1+1
17	Substitute: $2(5)^2 - 3(5) + 4$ (1); $= 39$ (1).	1+1
18	1 day = 86,400 seconds (1); $30 \times 86400 = 25,92,000$; international: 2,592,000 (1).	1+1
19	Innermost brackets; Next level (1); Final operations; Value $= 82$ (1).	1+1
20	The missing number is the product divided by the known factor (1); $\square = 3.6 \div 1.5 = 2.4$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$1.5 \times 0.6 = 0.9$ (1); $2.4 \div 0.6 = 4$ (1); Sum $= 4.9$ (1).	1+1+1
22	$P = 2[(7x + 3) + (2x - 1)]$; $= 2(9x + 2)$ (1); $= 18x + 4$ (1); At $x = 5$: 94 cm (1).	1+1+1
23	Compare numbers digit by digit (1); Descending: 98,07,654, 90,87,654, 90,78,654 (1); Difference $= 7,29,000$ (1).	1+1+1
24	First $= 252$ (1); Second $= (14 + 3) \times (18 - 3) = 255$ (1); Difference $= 3$ (1).	1+1+1
	OR: They separate by $4.2 + 3.8 = 8$ km every hour (1); After 1.5 hours: $8 \times 1.5 = 12$ km (1); Time $= 32 \div 8 = 4$ hours (1).	1+1+1
25	Numbers: $n, n + 1, \ldots, n + 3$ (1); Sum $= 4n + 6$ (1); At $n = 5$: 26 (1).	1+1+1
	OR: Distance $= 300000 \times 60 = 1,80,00,000$ km (1); In words: one crore eighty lakh (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Monthly $= 4,20,000 \times 150 = 6,30,00,000$ (1); Yearly $= 75,60,00,000$ (1); Words: seventy-five crore sixty lakh (1); Nearest crore: $76,00,00,000$ (1); Show working with place values (1).	1+1+1+1+1
	OR: Income $= 50 \times 4 \times 12 + 30 \times 3 \times 20$ (1); $= 4200$ (1); Expenses $= 700 + 500 = 1200$ (1); Profit $= 3000$ (1); Expression uses brackets correctly (1).	1+1+1+1+1
27	Net gain per minute $= 9.5 - 2.5 = 7$ litres (2); Time $= 84 \div 7 = 12$ minutes (3).	2+3
	OR: (i) $5(12) + 2 = 62$ (1); (ii) Figure 1 $= 7$ and figure 2 $= 12$ (1); Total $= 19$ (1); (iii) $62 - 57 = 5$ more tiles (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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